Software Engineering

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

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Section : G

Project : Domain & SSL Certificate Expiry Alert System

Epic 1: Domain & SSL Monitoring

Description:
Enable automated daily monitoring of SSL/TLS certificates and domain registrations to determine their expiry status and identify assets approaching expiration.

User Stories:

- SLP-5 – Perform SSL/TLS Certificate Checks — 8 points
- SLP-6 – Perform Domain WHOIS Checks — 8 points
- SLP-7 – Calculate Expiry Risk — 3 points

Total Story Points: 19

Epic 2: Expiry Alert & Escalation

Description:
Generate and deliver expiry alerts through configurable notification channels while escalating alerts to appropriate stakeholders as the expiry deadline approaches.

User Stories:

- SLP-8 – Trigger Expiry Alerts — 5 points
- SLP-9 – Configure Notification Channels — 5 points
- SLP-10 – Escalate Critical Alerts — 8 points

Total Story Points: 18

Epic 3: Alert Acknowledgement & Audit

Description:
Enable authorized stakeholders to acknowledge active alerts while maintaining an audit trail of alert handling and preventing unnecessary repeated notifications.

User Stories:

- SLP-11 – Acknowledge Active Alerts — 3 points
- SLP-12 – Record Alert Acknowledgement — 2 points
- SLP-13 – Suppress Acknowledged Alerts — 3 points

Total Story Points: 8

Epic 4: Compliance Dashboard & Reporting

Description:
Provide SysAdmins and Security Officers with dashboard and reporting capabilities to monitor the compliance and expiry status of all monitored domains and certificates.

User Stories:

- SLP-14 – View Asset Compliance Status — 5 points
- SLP-15 – Categorize Expiry Risk — 3 points
- SLP-16 – Generate Compliance Reports — 5 points

Total Story Points: 13

Sprint Execution

A one-week Sprint 1 was created in Jira with five selected user stories covering domain and SSL monitoring and expiry alert functionality. The sprint was started and the selected stories were progressed through the Agile workflow from To **Do → In Progress → Done**. All five planned stories were completed, resulting in 29 completed story points. A second one-week sprint was then used to complete the remaining alert management, acknowledgement, audit, and compliance functionality.

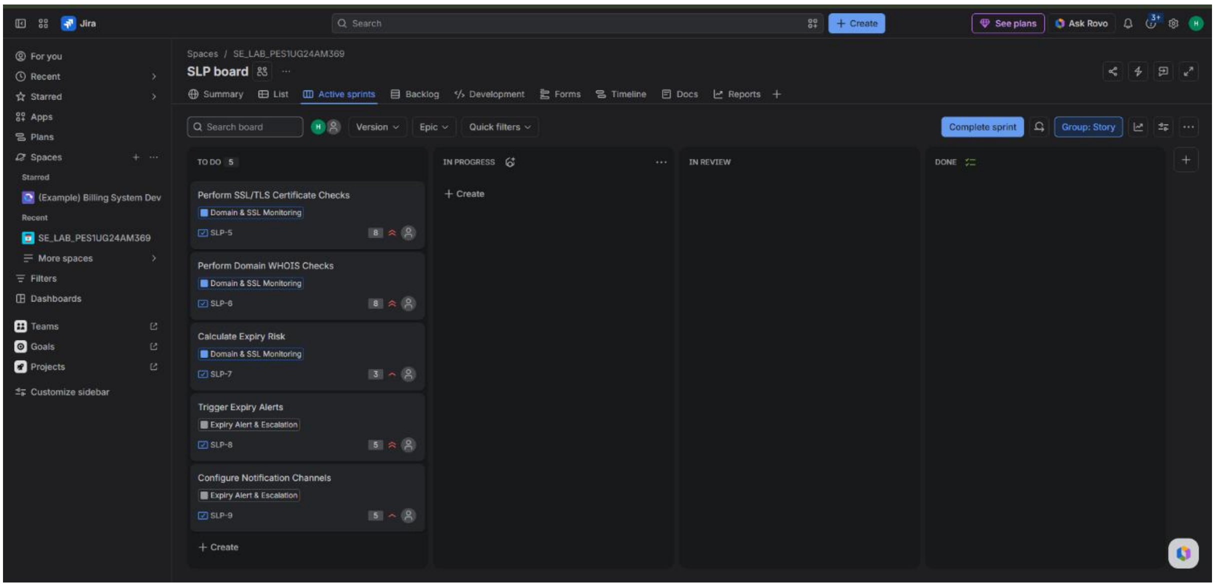
Story Point Estimation

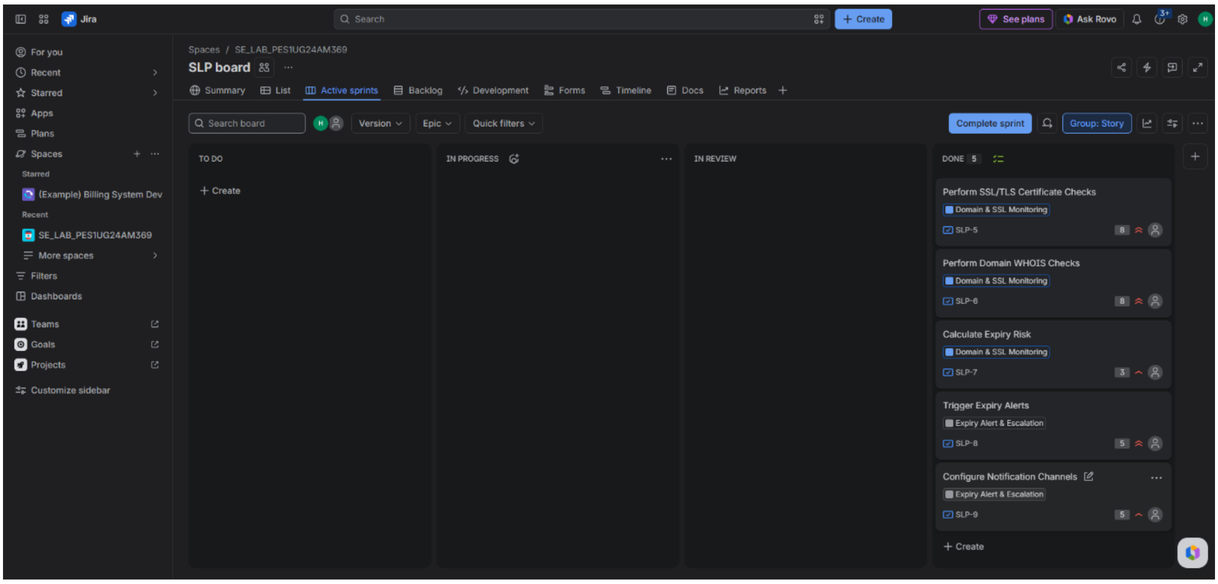
The lab requires assigning story points using the Fibonacci sequence. The story points were estimated based on the relative effort, complexity and uncertainty associated with each user story.

Story Points :

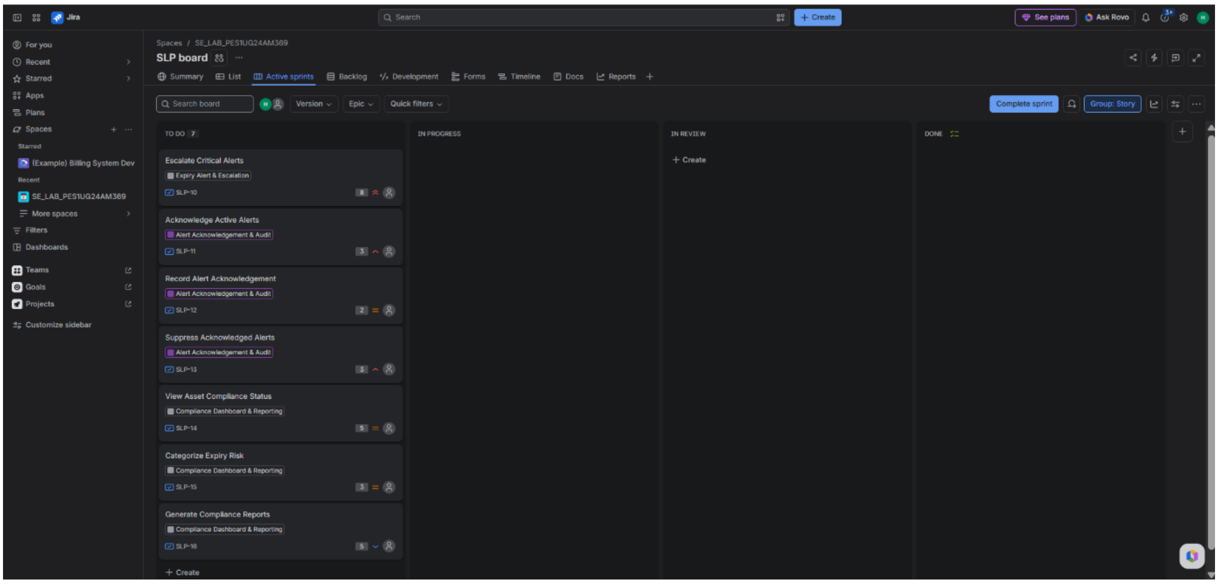
Story	**Story Points**
SLP-5 – Perform SSL/TLS Certificate Checks	8
SLP-6 – Perform Domain WHOIS Checks	8
SLP-7 – Calculate Expiry Risk	3
SLP-8 – Trigger Expiry Alerts	5
SLP-9 – Configure Notification Channels	5
SLP-10 – Escalate Critical Alerts	8
SLP-11 – Acknowledge Active Alerts	3
SLP-12 – Record Alert Acknowledgement	2
SLP-13 – Suppress Acknowledged Alerts	3
SLP-14 – View Asset Compliance Status	5
SLP-15 – Categorize Expiry Risk	3
SLP-16 – Generate Compliance Reports	5
Total	**58**

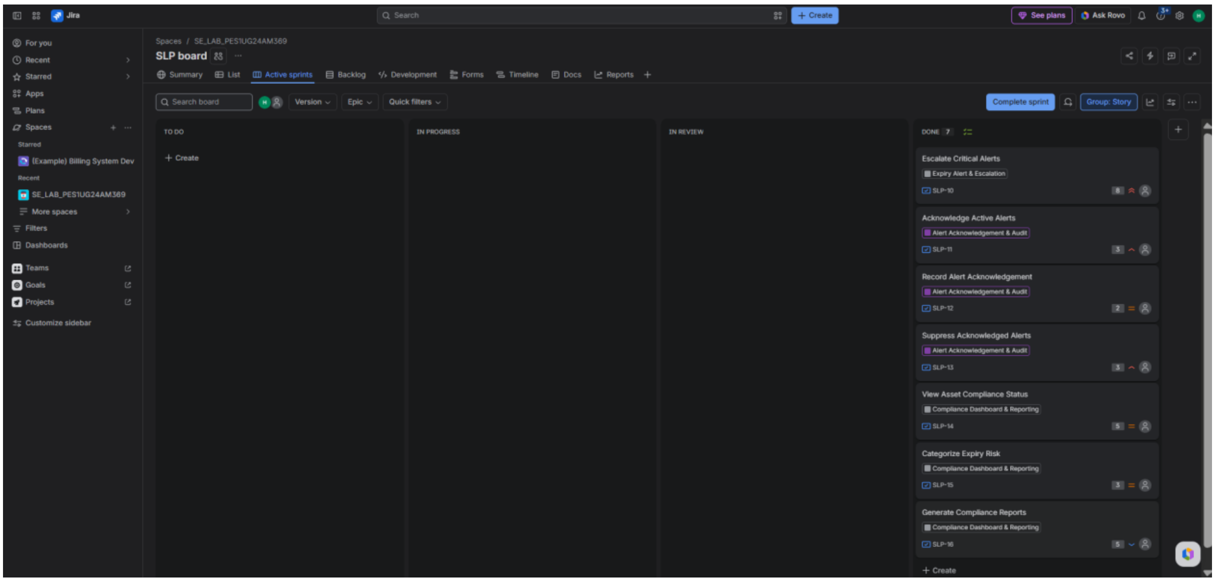
JIRA Sprint 1 Completion Screenshot :



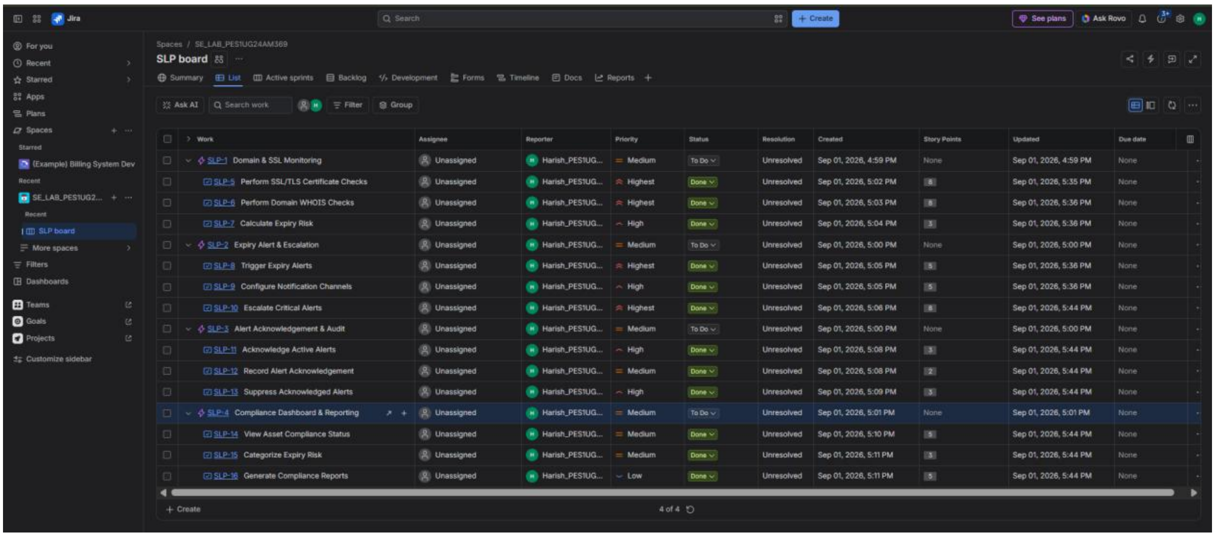


JIRA Sprint 2 Completion Screenshot :



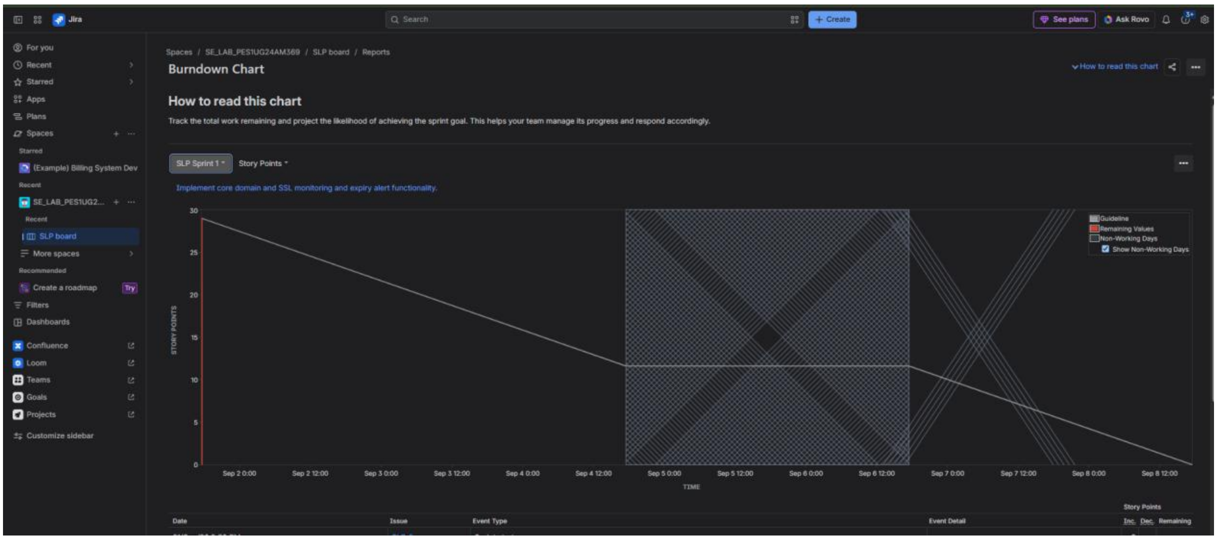


JIRA All Epics and Stories with Story Points :

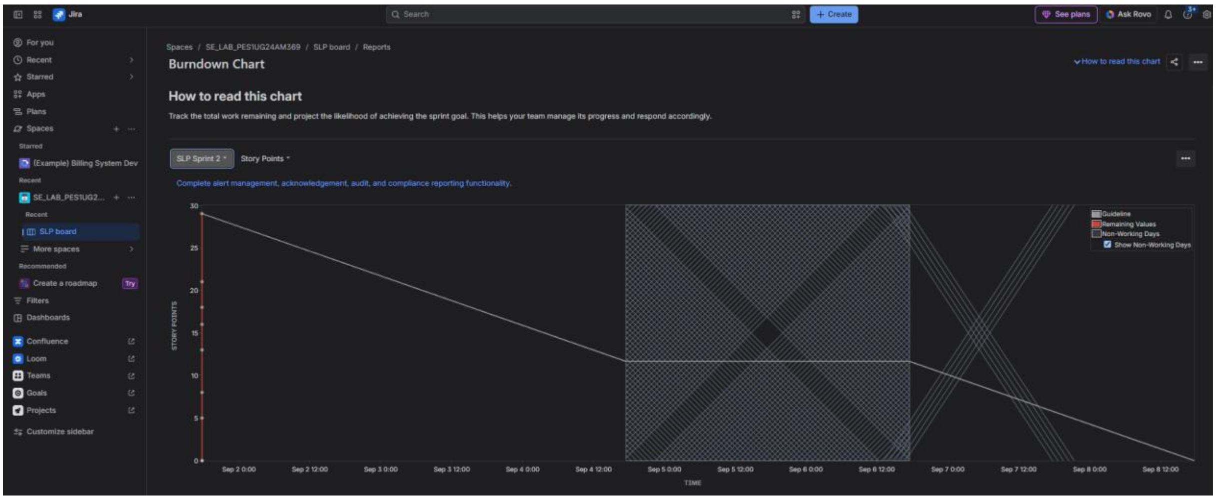


Burndown Chart :

SLP Sprint 1 :-



SLP Sprint 2 :-



Reflection Questions :

1. Did your estimations reflect the actual effort?

Ans : Yes, the story point estimations provided a reasonable representation of the effort and complexity involved in completing the user stories. Core monitoring activities such as SSL/TLS certificate checks and WHOIS checks were assigned higher points because they involve external checks and processing. Smaller tasks such as recording alert acknowledgements and calculating expiry risk required fewer points. The Fibonacci scale helped represent the increasing complexity and uncertainty of the different tasks.

2. Was your backlog well-prioritized?

Ans : Yes, the backlog was prioritized according to the importance of the functionality to the Domain & SSL Certificate Expiry Alert System. Core monitoring and alerting functionality was given higher priority because detecting certificate and domain expiry and notifying the responsible users are the main objectives of the system. Alert acknowledgement, audit functionality, dashboard visibility and reporting were prioritized subsequently as supporting functionality.

3. How did your simulated sprint align with your plan?

Ans : The simulated sprints aligned well with the planned backlog. Sprint 1 focused on core domain and SSL monitoring and expiry alert functionality, while Sprint 2 focused on alert escalation, acknowledgement, audit and compliance reporting. The selected user stories were progressed through the Agile workflow from To Do to In Progress and finally Done. Both sprints were completed successfully, covering all 58 planned story points.

4. What insights did the burndown chart give about your team's capacity?

Ans : The burndown chart provided a visual representation of how the remaining story points changed as work was completed during the sprint. It helped show the team's progress against the planned workload and provided an indication of whether the sprint work could be completed within the planned duration. The completed stories in the two sprints also demonstrated the importance of realistic story point estimation and balanced sprint planning.